

EXECUTIVE SUMMARY REPORT

Project: Netflix Dataset Analysis Author: Sumaiya (Data Science Intern) Date: July 2026 Status: Completed

SECTION 1: DATA CLEANING & CHALLENGES ENCOUNTERED

- **Missing Value Imputation (NaN):** The *director* and *cast* columns had heavy missing values, which were filled with placeholder strings like "Unknown Director" and "Unknown Cast" to preserve row integrity without dropping critical rows.
- **Date Standardization:** Inconsistent raw string dates in *date_added* were successfully parsed into unified *Datetime* format, allowing the clean extraction of a *year_added* column for trend mapping.
- **Multi-Value Splitting:** The comma-separated values in the *listed_in* (genres) column were processed using split and explode methods to fully isolate individual genres for precise analysis.

SECTION 2: KEY EDA FINDINGS & VISUALIZATIONS

- **Content Type Split:** A flawless **Pie Chart Analysis** reveals that **Movies dominate Netflix at 69.6%** of the catalog, while **TV Shows hold the remaining 30.4%**.
- **Timeline Distribution:** The left-skewed **Histogram (with KDE curve)** mathematically proves that the vast majority of content currently hosted on the platform was uploaded post-2015.
- **Statistical Outlier Analysis:** The **Boxplot Analysis** shows highly compact boxes between 2015-2021 (modern core), but features a long tail of individual outlier dots stretching back to the 1940s, showcasing Netflix's vintage film archive.

SECTION 3: TOP ACTIONABLE INSIGHTS

- **1. The Modern Strategy Pivot:** Netflix's explosive growth post-2015 confirms a permanent business shift toward prioritizing fresh, high-volume modern content releases and original productions over historical media.
- **2. Feature Films as the Core Product:** Despite the cultural popularity of binge-worthy TV series, Movies remain the structural foundation of Netflix's library, outnumbering TV shows by a clear 2:1 ratio.
- **3. Global Hyper-Localization:** "International Movies" emerged as the absolute #1 top genre in the Bar Chart analysis, proving that Netflix's aggressive global expansion relies entirely on localized regional content.

FINAL DELIVERABLES PACKAGE:

- `Cleaned_Netflix_Titles.csv` — Cleaned and preprocessed dataset ready for modeling.
- Comprehensive Jupyter Notebook — Complete workflow with Dataset Understanding, Visualizations, and Clean Code.
- One-Page PDF Summary Report — Official summary documentation (This Document).